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Traffic Animation of Pedestrian and Vehicle Interactions in Non-Signalized Mid-Block Regions

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ABSTRACT Traffic animation enables the visualization of potential conflicts among road users, supporting the development of safety measures and serving applications in education and entertainment. This paper presents a factor-based approach to generate animations of pedestrians interacting with vehicles in mid-block crossing scenarios. A mid-block refers to a road segment between intersections without traffic signals. In these settings, vehicles and pedestrians should coordinate dynamically to avoid accidents as there is a lack of designated crossing control. Furthermore, this dynamic environment provides an opportunity to produce traffic animation that depicts the diverse and intricate interactions between vehicles and pedestrians. We propose a dynamic graph that tracks leader-follower relationships among vehicles, as well as the evolving gap regions between them. In addition, pedestrians evaluate these gap regions using five intuitive factors, which are derived from gap characteristics. These factors are adjusted region distance, degree of passage, region growth rate, region flow ratio, and vacancy ratio. By tuning the influences of these factors, animators can simulate a wide range of pedestrians behaviors, from cautious waiting to assertive movement. To evaluate the performance of the proposed approach, we conducted experiments assessing effects of these factors on route decisions and pedestrian interaction styles. Additionally, we analyzed the impact of varying pedestrian and vehicle volumes on average travel time and distance. The results demonstrated the system's ability to produce plausible animations of pedestrian-vehicle interactions. Our system serves as a customizable tool for animators to create diverse pedestrian crossing behaviors tailored to dynamic vehicle gap scenarios. Our approach could be applied to various domains, such as education, training, and immersive entertainment. In these areas, tailored pedestrian crossing behaviors enhance the richness and engagement of traffic scenarios.

INDEX TERMS Dynamic graph, mixed-traffic simulation, pedestrian-vehicle interaction, local route decision making.

I. INTRODUCTION

Traffic animation has been applied in a wide range of applications, including self-driving systems [1], driving training systems [2], virtual reality [3], and education [4]. For these contexts, high-fidelity animation is essential for training, testing, and system evaluation. This is especially important for modeling dynamic interactions, such as those between vehicles or between vehicles and pedestrians. Furthermore, complex or critical behaviors, such as pedestrians jaywalking or suddenly rushing in front of vehicles, are often challenging to collect in real-world scenarios due to their rarity and unpredictability [5] [6]. These behaviors may stem from distractions, emergencies, or misjudging movement [7]. Therefore, developing tools that support the creation of desirable and diverse pedestrian behaviors in pedestrian-vehicle interaction animations is essential. To address this gap, our system provides animators to design diverse pedestrian crossing behaviors in midblock areas, which are road segments located between two intersections and typically do not have traffic signals. Focusing on midblocks is important due to the complex and unregulated interactions with vehicles. This setting offers an opportunity to use traffic animation to illustrate pedestrian-vehicle dynamics. Our study focuses on generating animations of such interactions, where pedestrians may exhibit conservative or risky crossing behaviors.

Various topics have been investigated for vehicle simulation [8], such as car-following models [9]–[11], continuous lane changes [12], and mixed traffic [13], [14]. Pedestrian

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simulation techniques consider diverse behavioral, environmental, personality, and spatial factors, including right of way [15], emotion [16], and competing behaviors in evacuation [17], [18]. However, the aforementioned techniques simulate pedestrians and vehicles using one-way interaction, where each behaves independently without considering the others' actions.

When pedestrians and vehicles enter a neighborhood, their behavior can influence each other [19]. In signalized regions, *right of way* is clear to road users. This allows road users to use the road alternately in a safe manner [20]. However, in special situations or non-signalized areas, road users may use the road even though *right of way* is not theirs at the moment [21]. In such situations, the pedestrians and vehicles involved should coordinate with each other to avoid collisions. For example, they may adjust and yield to the reactions of others. However, some road users, such as children or distracted drivers, may not perform actions properly and other road users misinterpret their intentions [22] [23]. Therefore, road users must make careful decisions in such situations to avoid accidents.

In general, pedestrians may not cross roads in certain non-signalized road segments because the traffic flow can be affected adversely and accidents may occur. This is because both vehicles and pedestrians use the roads at the same time. However, on rainy days, traffic congestion, and emergent situations, the vehicles may move slowly and the pedestrians tend to walk across the road to another side. In addition, people are more likely to cross directly or do not obey rules if there is a longer walk to a pedestrian bridge or signalized crosswalk [24], [25]. In addition, there are various crossing behaviors of pedestrians [21], [26], including one-stage, two-stage, perpendicular direction, oblique direction, and three-stage. There are wait times and safety concerns with different pedestrian crossing behaviors. Pedestrian decision-making is influenced by several factors, including their perception of the traffic environment, individual temperament (e.g., preferred walking speed and caution level) [27], and *time to collision* (TTC) with vehicles [28]. Demographic traits like gender and age also affect crossing behavior [29]. TTC refers to the estimated time remaining before a collision would occur, assuming both the pedestrian and vehicle maintain their current velocities. It is a crucial metric for ensuring effective collision avoidance. Furthermore, gap acceptance is an essential factor when pedestrians make a decision based on their waiting time and the distance from approaching vehicles [30]. Thus, a decision model based on gap acceptance is required to simulate the process of pedestrians making decision when encountering vehicles [31].

To couple the interactions between road users (i.e., pedestrians and vehicles), their influences should be considered as a whole. Cellular automaton (CA) methods [32]–[34] can efficiently simulate the interaction between pedestrians and vehicles in non-signalized regions. They enable fast updates of pedestrian and vehicle positions due to their discrete simulation space. However, CA methods cannot be directly applied to traffic animation because the motion of road users (e.g., pedestrians and vehicles) is updated in a discrete manner, i.e., jumping from cell to cell. In addition, CA methods cannot reproduce the detailed interaction between pedestrians and vehicles, such as the changes in pedestrian and vehicle movement speeds along the lateral and longitudinal directions. As a result, the relative positions between vehicles and pedestrians are not easy to simulate adequately. Consequently, force-based methods aim to simulate continuous interactions among heterogeneous road users, such as vehicles, pedestrians, or bicycles [35]–[37]. These methods incorporate pedestrian decision-making to some extent and primarily focus on local collision avoidance. However, they do not explicitly consider the characteristics of gap regions, which refer to the space between a leading vehicle and a following vehicle in a leader–follower relationship. As a result, these methods have limited ability to model the nuanced behaviors that pedestrians exhibit when choosing and crossing vehicle gaps.

In this study, we proposed a factor-based method to produce animations of pedestrians interacting with vehicles (e.g., cars, buses, and motorcycles) in mid-block areas. While pedestrians cross road between vehicles, they select gap regions to cross. They rely on the characteristics of these gap regions to make route decisions. These characteristics are relevant to the relative speeds of vehicles, growth rates of gap regions, directions of pedestrian movements, and pedestrian densities. As vehicles move dynamically, their relative positions change continuously, leading to evolving gap topologies. Therefore, a static representation for the gap topologies would be insufficient. Therefore, we employ a dynamic graph structure that captures both the spatial relationships and physical characteristics of gap regions, reflecting changes in vehicle positions and the corresponding gap regions. However, pedestrians' visual perception is often constrained by nearby vehicles, which may obstruct their field of view and limit their awareness to surroundings [38]. Moreover, the continuous movement of vehicles leads to dynamic and complex traffic patterns [39]. Therefore, we focus on a local decision model that accounts for surrounding vehicles and pedestrians. As vehicles move, they establish leader-follower relationships, which are dynamically tracked using the dynamic graph structure that monitors the gaps between each pair. Pedestrians make decisions based on five behavioral factors derived from the characteristics of these gaps. The factors are *Adjusted Region Distance*, *Degree of Passage*, *Region Growth Rate*, *Region Flow Ratio*, and *Vacancy Ratio*. By adjusting the influence of these intuitive factors, the system enables animators to generate diverse pedestrian-vehicle interaction scenarios. To produce animations, we implement the system for generating pedestrian and vehicle trajectories, which are then visually rendered using Unity. The system supports various crossing behaviors, including cautious waiting, group movement, avoidance of crowded areas, and risky crossings.

Our contributions include: (1) Propose a dynamic graph that dynamically tracks the leader-follower relationship of ve-

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hicles and gap regions formed by the leaders and corresponding followers. To do so, road lanes are partitioned into layers to support fine-grained pedestrian navigation. Thus, the graph gives pedestrians real-time access to the characteristics of the gap region across layers, which supports them in making informed and adaptive crossing decisions. (2) Propose five behavioral factors derived from the characteristics of gap regions to guide pedestrian decision-making in midblock crossing scenarios. They are designed to be intuitive, making them easy to interpret and adjust.

Our system is a flexible tool for animators, which allows them to adjust the five behavioral factors to shape pedestrian behavior. The system helps create animations that may support safety visualization, public education, and synthetic data generation. Visualizing these interactions could help illustrate pedestrian safety, raise public awareness, and contribute to engaging simulation or entertainment scenarios.

II. RELATED WORK

This section reviews the techniques for pedestrian simulation, vehicle simulation, and pedestrian-vehicle interactions.

Pedestrian Simulation Techniques: Cellular automaton models [40], [41] represent the simulation space as regular grid cells. The motion of a road user is depicted in a discrete manner, moving from cell to cell. The social force model [42] is a typical representative of continuous models, which drive the motion of pedestrians based on forces that are induced by surrounding objects and destinations. There are a large number of variant models that can simulate various behaviors of pedestrians [43]–[45].

Vehicle Simulation Techniques: Vehicle simulation involves modeling the behavior of vehicles and their interactions across various driving scenarios. The primary aspects include car-following and mixed traffic interactions. Car-following models simulate the interaction between successive vehicles [9], [46]. In general, these models use mathematical equations or rules to describe the vehicle behaviors. For example, collision avoidance models consider the safe following distance and the safety reaction time [47]. The action point models are based on the driver's perception of the involved vehicles [48]. The fuzzy logic-based models represent the vehicle behavior as a logic system with uncertainty [49]. The stimulus-response models describe the response of the following vehicles based on the leader vehicles [11]. In this paper, we adopt a stimulus-response model to simulate car-following behavior.

Cellular Automaton-Based Interactions: Cellular automaton models are effective in modeling vehicle yielding and pedestrian crossing behaviors [50]. In general, microscopic update rules are used to update decisions and positions of pedestrians and vehicles. For safety considerations, pedestrians rely on different criteria of acceptable gaps to make decisions [51]. These models are calibrated with real data and field observations to reproduce realistic interactions between pedestrians and vehicles [33]. A bidirectional pedestrian model can be used to simulate the pedestrians moving in counter-directions in a deadlock-free manner. The perception and judgment of pedestrians and drivers are handled by an evolutionary game-based decision model [32]. Neighboring pedestrians tend to follow a pedestrian who begins to cross a road due to the herd mentality [52].

Force-Based Interactions: A significant number of social force-based techniques, inspired by [42], have been developed to continuously simulate the interaction behaviors of pedestrians and vehicles. Primarily, social force models describe how pedestrians and vehicles generate repulsive and attractive forces that influence each other. Subsequently, the accelerations, velocities, and positions of the pedestrians and vehicles are updated based on the forces exerted on them.

To model the interaction between pedestrians and vehicles, pedestrians perform a decision-making process to determine their actions (e.g., cross or wait) based on a gap acceptance criterion. This criterion considers factors like available space and driving styles (e.g., aggressive or conservative) of oncoming vehicles. In addition, simulation models for both pedestrians and vehicles take into account environmental feedback mechanisms [31]. The motion of an agent (such as a vehicle, bicycle, or pedestrian) is influenced by various factors, including the agent's desire, neighboring road users, and road structures. Such factors can be converted to forces that are used to drive the motion of the agent [36], [53]. In addition, based on real-driving data, optimal parameters for a vehicle dynamic model can be calibrated to simulate vehicles interacting with pedestrians [54].

At signalized intersections, pedestrian behavior is affected by group or individual evasion with other pedestrians, collision avoidance with incoming vehicles, road signals, and crosswalk boundaries. These factors can be represented as social forces that are incorporated with a pedestrian psychological process to reproduce trajectories of pedestrians [55]. In relatively free space, the shapes and longitudinal velocities of vehicles have a great impact on the movement of pedestrians [35], [37]. Drivers and pedestrians have different judgement behaviors, including risk perception, gap judgement, crosswalk judgement, and distance judgement. Probability models have been adopted in car-following models [34] to model such behaviors. In addition, a mixed model can be adopted [56], such as using a CA component for discrete pedestrian simulation and a modified Gipps car-following model for continuous vehicle simulation.

Deep Learning Techniques for Pedestrians and Vehicles: Deep learning leverages artificial neural networks to perform complex tasks and has been adopted in pedestrian-related research. For instance, Convolutional Neural Networks (CNNs) have been used to detect vehicles and pedestrians in urban environments [57], while pedestrian trajectory prediction has been addressed using both learning-based and classical knowledge-based models in sparse and crowded settings [58]. Beyond perception and prediction, deep learning techniques have also been applied to assess the severity of pedestrian-vehicle interactions at unsignalized intersections [59]. To model social dynamics, deep inverse reinforcement

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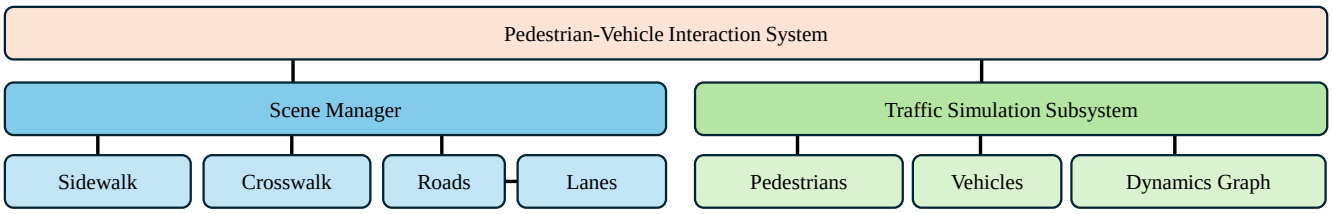


FIGURE 1. System architecture.

learning has been employed to capture pedestrian-pedestrian interactions [60]. LSTM (Long Short-Term Memory) models integrated with reinforcement learning have guided pedestrian navigation in unfamiliar environments by avoiding static and dynamic obstacles [61]. In robotic applications, deep reinforcement learning enables proactive collision avoidance with pedestrians in crowded spaces [62]. Furthermore, in mixed traffic scenarios, a deep reinforcement learning-based approach has been developed to simulate pedestrians' risk-awareness of oncoming vehicles by incorporating post-encroachment time metrics [63]. Collectively, these methods contribute to either predicting pedestrian trajectories or simulating pedestrian behaviors in dynamic, interactive environments. However, they often lack mechanisms for designing crossing behaviors that respond to the spatial and temporal characteristics of gap regions. To address this, we introduce a framework that enables animators to modulate the influence of five behavioral factors, facilitating the generation of diverse and intentional pedestrian actions in pedestrian-vehicle interaction scenarios.

III. SYSTEM OVERVIEW

Our system consists of a scene manager and a traffic simulation subsystem (see Figure 1). The scene manager maintains road structures and road connectivity information. The traffic simulation subsystem simulates the movements of vehicles and pedestrians. It also maintains a dynamic graph that is used to simulate pedestrian-vehicle interactions in mid-block regions. The attributes of an agent (a vehicle or a pedestrian) include its dimensions, position, velocity, and desired speed. Vehicles move in the longitudinal direction along the x-axis, while the lateral direction is represented by the y-axis. During the simulation, pedestrians are modeled to cross roads to reach their destinations. Pedestrians choose gaps between vehicles based on their preferences and ensure these gaps avoid collisions. Since the gaps between the vehicles change dynamically, the pedestrians make these gap selections locally. Vehicles are simulated based on a car-following model. When interacting with pedestrians, vehicles adopt strategies to avoid collisions. The simulation is finished when all pedestrians have reached their destinations. Then the system generates animations using the simulated trajectories of the pedestrians and vehicles.

IV. DYNAMIC GRAPH

A. THEORETICAL BACKGROUND

Pedestrians perceive gap regions between vehicles and assess their characteristics to make crossing decisions. To assist such decisions, a dynamic graph is constructed to track adjacent vehicles with leader-follower relationships and their associated gap regions. The dynamic graph is updated based on the positions of vehicles. If two vehicles do not change the leading and following relationship, these two vehicles maintain the same gap but only the characteristics associated with the gap may change over time. So when the vehicles change the leading and following relationship, we only have to deal with the newly created gaps, or the gaps no longer exist. For example, when a vehicle moves from one lane to another, new gaps are created and existing gaps are affected by the vehicle. For the other gaps that are not formed by the vehicle, these gaps are not affected. Thus, we can update the dynamic graph incrementally based on the vehicles that change lanes. The dynamic graph also keeps track of the connectivity of adjacent gaps, which is useful for simulating pedestrians moving from one gap to another.

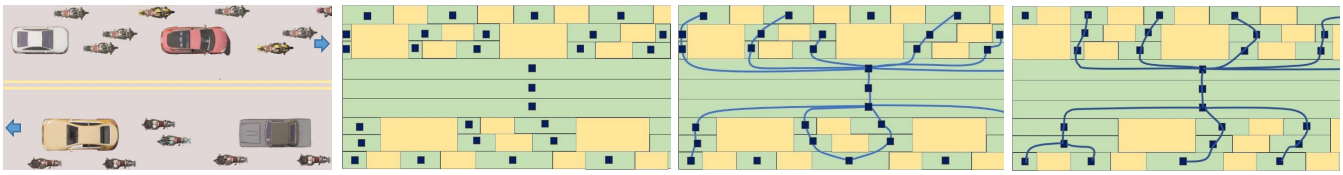
B. DYNAMIC GRAPH CONSTRUCTION AND UPDATE

We divide a road segment into a set of layers. A gap region is defined as the space between the rear end of the leading vehicle and the front end of the following vehicle, spanning the full width of the layer. If no vehicles are present, the gap region corresponds to the entire layer. The width of a layer is set to a value not larger than the minimum vehicle width to accurately represent pedestrians' moving areas along the layer. In our case, the width of a layer is approximately the width of a motorcycle. Figures 2(a,b) illustrate a road segment with left-hand traffic on a two-lane road and the layers of the corresponding road segment.

A dynamic graph $G = (V, E)$ consists of a set of nodes (or vertices) V and a set of edges E. A node represents a gap region formed by a leading vehicle and its corresponding follower in a leader–follower relationship. Each node stores the pair of the leading and following vehicles. Two gap regions are adjacent if they overlap or have common boundaries. An edge connects two adjacent gap regions. The edge indicates that a pedestrian may walk from one gap region to another. Thus, the edges represent the connectivity of gap regions on adjacent layers. Figure 2(c) illustrates a dynamic graph. As the vehicles move, adjacent gap regions in the previous

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(a) A two-lane road. The arrows indicate the movement directions.

(b) The road is divided into layers bounded by the horizontal line segments.

(c) Edges connect gap regions that are adjacent to each other.

(d) The dynamic graph is updated after the vehicles change positions.

FIGURE 2. Dynamic graph: structure and update. Green rectangles denote gap regions, yellow rectangles indicate vehicles, and blue squares represent the nodes associated with each gap region. Edges connect corresponding pairs of nodes.

simulation frame may not be adjacent in the current frame. In this case, the corresponding edges connecting the pairs of non-adjacent gap regions are removed. Figure 2(d) shows the dynamic graph after the positions of the vehicles are updated.

We construct a dynamic graph of a road segment as follows. The road segment is divided into a set of layers along the lateral direction. The set of layers is denoted as an order set $L = \{L_1, L_2, \cdots, L_n\}$, where n is the number of layers, and L_i is a layer with index i, where $1 \leq i \leq n$. As we simulate the movement of each vehicle, we know which layers the vehicle overlaps based on the vehicle's dimension and position. The vehicle is on a layer *if and only if* it overlaps with the layer. It can occupy one or more layers. On a layer, two vehicles form a leader-follower relationship *if and only if* one vehicle is in front of another vehicle and there are no other vehicles between them. The former is called the leading vehicle (or leader) while the latter is the following vehicle (or follower). A node is created for these two vehicles, representing the gap region between the two vehicles. Because the vehicles are ordered on a layer, the gap regions on a layer are also ordered. After constructing the nodes, we proceed to construct the edges for the adjacent gap regions. For each adjacent layer pair (L_i, L_j), we examine every node pair (n_i, n_j), where n_i belongs to L_i and n_j belongs to L_j. If the corresponding gap regions are adjacent, an edge is created between n_i and n_j.

During the simulation process, the positions of vehicles and pedestrians are updated in every simulation frame. The dynamic graph is then updated to ensure the integrity of nodes and edges. A node is retained if the corresponding two vehicles maintain the leading-following relationship. However, a vehicle may move from its current layer to another layer (e.g., changing a lane). In this case, nodes are created for the new gap regions. In addition, if the previous gap regions no longer exist in the current simulation frame, the corresponding nodes are removed. Similarly, new edges are created for new pairs of adjacent gap regions. However, the edges will be removed if the corresponding nodes no longer exist or are no longer adjacent to each other.

V. LOCAL ROUTE DECISION FOR PEDESTRIANS

We aim to simulate pedestrian interacting with vehicles through plausible behaviors, reflecting varied preferences in gap selection criteria. As pedestrians cross the road, they sequentially choose gap regions to transition between layers. This process continues until they reach the opposite side. We assume that pedestrians act with the intent to avoid accidents. They select collision-free gap regions based on factors such as time-to-collision and vehicles' yielding behaviors. However, they may still make risky crossing decisions that require vehicles to brake sharply to prevent collisions. Modeling these interactions is essential for animators creating physically plausible and dynamic scenarios.

A pedestrian selects a gap region based on five behavioral factors: (I) *Adjusted Region Distance*, (II) *Degree of Passage*, (III) *Region Growth Rate*, (IV) *Region Flow Ratio*, and (V) *Vacancy Ratio*. These factors are derived based on the characteristics of gap regions (see Figure 3). They are designed to have higher values when pedestrians are more likely to select the corresponding gap regions. However, a negative contribution of a factor may incentivize pedestrians to pass through a gap region that exhibits a correspondingly low value for that factor. Consider a pedestrian i at a gap region k. Let $L^k(i)$ be the set of gap regions in the next layer connected to gap region k. Thus, the pedestrian i will choose one of the gap regions in $L^k(i)$ as a temporary destination. For each gap region $g \in L^k(i)$, a value is associated with each factor, which is calculated based on all the adjacent gap regions of k in $L^k(i)$. For each factor, we consider the influence of all gap regions as a whole to determine how much a gap region affects the pedestrian. The details for calculating the five factors are provided below.

1) *Adjusted Region Distance:* Pedestrians tend to choose closer gap regions for minimizing their travel distance and physical effort. Denote $d(\mathbf{p}_i, g)$ as the distance between the pedestrian i and a gap region g. The *Adjusted Region Distance* for pedestrian i, measured from the gap region k to the gap region g, is defined as:

$$\sigma_1^{k\to g} = 1 - \frac{d(\mathbf{p}_i, g)}{\sum_{g' \in L^k(i)} d(\mathbf{p}_i, g')}. \tag{1}$$

Thus, the gap region g with the smallest distance $d(\mathbf{p}_i, g)$ will be assigned the highest *Adjusted Region Distance* $\sigma_1^{k\to g}$.

2) *Degree of Passage:* It indicates how easy the pedestrian i can move from the gap region k to a target region g. The *Degree of Passage* depends on how much the two gap regions k and g overlap. However, a pedestrian cannot cross a gap region whose width is smaller than

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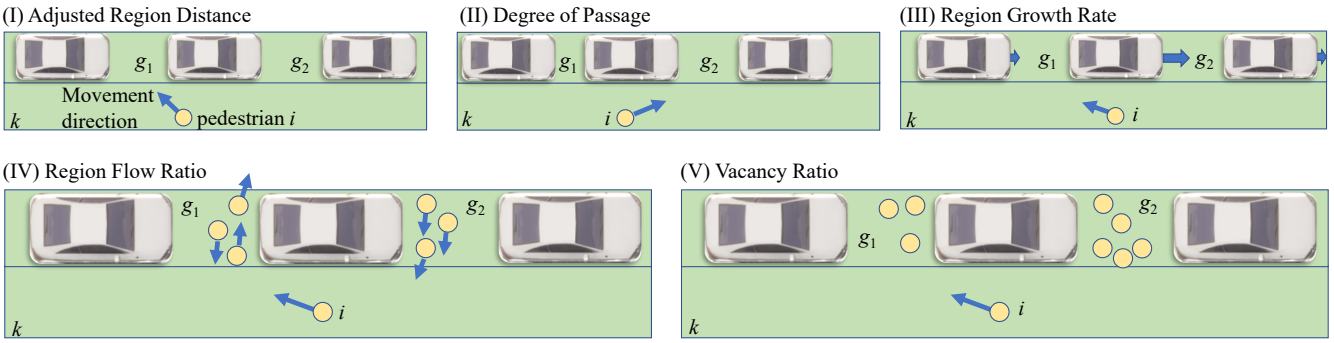


FIGURE 3. **Illustration of the five factors. Pedestrian *i* is located within gap region *k*, with an arrow indicating their preferred movement direction.**

the pedestrian size. Thus, $\eta_2^{k\to g}$ is calculated as:

$$\eta_2^{k\to g} = \max\left\{\left(\frac{\Delta x_{k\to g} - 2r_i}{2r_i}\right), 0\right\}, \tag{2}$$

where r_i is radius of the pedestrian i, and $\Delta x_{k\to g}$ is the length by which the regions k and g overlap. Thus, a greater overlap between regions k and g leads to a higher value of $\eta_2^{k\to g}$, indicating it easier for the pedestrian to transition from k to g. The *Degree of Passage* $\sigma_2^{k\to g}$ for gap region g is computed by:

$$\sigma_2^{k\to g} = \frac{\eta_2^{k\to g}}{\sum_{g'\in L^k(i)} \eta_2^{k\to g'}}. \tag{3}$$

If pedestrians want to avoid potential congestion or blockage, they may choose a gap region with a higher *Degree of Passage*. However, they might still take the risk of passing through a gap region with a lower *Degree of Passage*.

3) *Region Growth Rate:* It refers to how fast a gap region g expands along the longitudinal direction between the leading and following vehicles. Denote s_1^g and s_2^g as the leading and following vehicle speeds along the longitudinal direction, respectively. The speed difference between them is defined as:

$$\eta_3^g = s_1^g - s_2^g. \tag{4}$$

A positive η_3^g indicates the region is growing. The *Region Growth Rate* for g is computed by:

$$\sigma_3^{k\to g} = \frac{\eta_3^g + |\min(s_{\min}, 0)|}{\sum_{g'\in L^k(i)}(\eta_3^{g'} + |\min(s_{\min}, 0)|)}, \tag{5}$$

where $s_{\min}$ represents the minimum value of $\eta_3^{g'}$ across all gap regions g' within the layer $L^k(i)$. Pedestrians tend to cross a layer through gap regions that exhibit increasing width, facilitating easier passage. However, they may still attempt to navigate through narrower gaps with a low *Region Growth Rate* by positioning themselves between vehicles.

4) *Region Flow Ratio:* It is related to the ratio of pedestrians moving along a direction to the total number of pedestrians in a gap region. We take into account the bi-directional movements of the pedestrians crossing the road segment. For each region, we have two region flow ratios, one for each movement direction. We define

$$\eta_4^g = \begin{cases} 0.5, & \text{if } n^g = 0 \\ \frac{n^g(i)}{n^g}, & \text{otherwise,} \end{cases} \tag{6}$$

where n^g is the total number of pedestrians in a gap region g, and $n^g(i)$ is the number of pedestrians moving in the same direction as pedestrian i. The *Region Flow Ratio* of g is calculated by:

$$\sigma_4^{k\to g} = \frac{\eta_4^g}{\sum_{g'\in L^k(i)} \eta_4^{g'}}. \tag{7}$$

When the contribution of *Region Flow Ratio* is high, pedestrians tend to select regions where others move in the same direction and avoid those with opposing flow. In contrast, a negative contribution encourages pedestrians to enter regions moving against the dominant pedestrian direction.

5) *Vacancy Ratio:* This refers to how spacious the clearance area is within a gap region. It is derived from the occupancy ratio, η_5^g, which represents the proportion of the region's area occupied by pedestrians in gap region g. η_5^g is calculated as follows:

$$\eta_5^g = \max\left\{\frac{\tau_g^p}{\tau_g}, 1\right\}, \tag{8}$$

where τ_g^p is the total area occupied by pedestrians in g, and τ_g is the region area. The *Vacancy Ratio* for g is computed by:

$$\sigma_5^{k\to g} = 1 - \frac{\eta_5^g}{\sum_{g'\in L^k(i)} \eta_5^{g'}}. \tag{9}$$

Pedestrians are more likely to enter gap regions with a higher *Vacancy Ratio*, which indicates greater available space. However, pedestrians may also enter regions with a lower *Vacancy Ratio*, as these areas are perceived to offer greater safety due to the presence of others.

Examples: Figure 3 illustrates the five factors used to evaluate two gap regions. The set of gap regions that pedestrian

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i considers is $L^k(i) = \{g_1, g_2\}$. The following assumes that pedestrian i prefers gap regions associated with higher values of a given factor. (I) g_1 is preferred because its *Adjusted Region Distance* is greater than that of g_2, indicating it is closer to pedestrian i. (II) g_2 has a higher *Degree of Passage* than g_1 and thus, pedestrian i moves to region g_2. (III) Since the middle car is moving forward at a higher speed, g_1 expands, indicating a positive *RegionGrowthRate*, while g_2 contracts. (IV) In region g_1, the *Region Flow Ratio* of pedestrians moving in the same direction as pedestrian i is higher than that in region g_2. (V) The areas of g_1 and g_2 are the same, but region g_1 is preferred as its *Vacancy Ratio* is higher than that of g_2.

Degree of Preference: The degree of preference of a region indicates the tendency of the pedestrian i to select the region as the temporary destination. It is a linear weighted combination of the values associated with the five factors, which is computed by:

$$\sigma^{k\to g} = \sum_{j\in\{1,2,\cdots,5\}} w_j \sigma_j^{k\to g}, \tag{10}$$

where w_j are weights, which control the contributions of the factors to the decision-making of the pedestrian. A positive weight of a factor causes pedestrians to move to a gap region with a higher value of the corresponding factor, and otherwise causes pedestrians to move to a gap region with a lower value. The pedestrian chooses the region with the highest *Degree of Preference* as the temporary destination. If two or more regions have the same aggregate value, the pedestrian selects the closest region. To simulate stable pedestrian movement from one layer to the next, two successive gap regions are evaluated at a time. This enables the pedestrian to anticipate future movement and avoid unnecessary stops or sharp turns when encountering vehicles. In contrast, to simulate distracted pedestrians, gap regions are selected layer by layer, reflecting short-term decision-making behavior.

VI. SYSTEM IMPLEMENTATION

The simulation process has two stages: initialization and runtime. During initialization, the attributes of the vehicles and pedestrians are properly set. The system populates vehicles along the road, obtaining all pairs of leading and following vehicles. At the same time, a dynamic graph is constructed for the vehicles. During the runtime stage, vehicle movements are updated based on neighboring pedestrians and their leaders. Then, the dynamic graph is updated based on the vehicles' positions. In addition, the pedestrians are updated based on pedestrian models and the dynamic graph for route selection. This process repeats until the simulation is terminated.

A. PEDESTRIAN SIMULATION

When pedestrians cross a road, they move layer by layer, seeking gap regions in the next layer as temporary destinations. To guide their movement, an attractive force is applied at these destinations. If the gap regions are directly in front of them, the nearest points within those regions are selected as temporary targets, allowing pedestrians to walk straight toward them. However, if vehicles obstruct their path, the temporary destinations are repositioned to either the rear or front of the blocking vehicles. In such cases, pedestrians move longitudinally along the road to approach the gap regions until they are directly ahead, at which point they can enter the gaps and continue crossing. We adopt a force-based model to update the position of pedestrians [36] [31] to achieve collision avoidance. The net force acting on a pedestrian is determined based on the temporary destination, nearby vehicles, surrounding pedestrians, and environmental factors. Let i be the pedestrian and I be the set of the nearby pedestrians. Formally, the net force is calculated by

$$\mathbf{f}_i = \mathbf{f}_i^0 + \sum_{j\in(I-\{i\})} \mathbf{f}_{ij} + \sum_W \mathbf{f}_{iW} + \sum_U \mathbf{f}_{iU}, \tag{11}$$

where $\mathbf{f}_i^0$ drives the pedestrian to the temporary destination, $\mathbf{f}_{ij}$ is the interaction force between pedestrians i and j, $\mathbf{f}_{iW}$ is the repulsive force by neighboring road users, and $\mathbf{f}_{iU}$ is the influence force from an obstacle U in the environment. The forces are modeled as exponential decay functions that reduce their effect on pedestrians as the distance increases.

B. VEHICLE SIMULATION

We employ the intelligent driver model to simulate free driving and car-following behaviors of vehicles [64]. To achieve a desired speed v^* in free driving, the acceleration (a_{free}) of the subject vehicle is computed by

$$a_{\text{free}} = a_{\max}\left(1 - \left(\frac{v}{v^*}\right)^{\delta}\right), \tag{12}$$

where $a_{\max}$ is the maximum acceleration and v is the current speed of the subject vehicle. The acceleration factor δ influences how quickly the vehicle's speed changes towards the desired speed.

When a leading vehicle is in front of the subject vehicle, the subject vehicle may need to decelerate to keep a safe distance from the leading vehicle. Thus, the braking deceleration b depends on the comfortable deceleration b_{com}, current gap s, and the desired minimum gap s^* to the leading vehicle. Thus, we have

$$b = a_{\max}\left(\frac{s^*(v,\Delta v)}{\max(s, s_0)}\right)^2, \tag{13}$$

where the desired minimum gap is calculated by

$$s^*(v,\Delta v) = s_0 + vT + \frac{v\Delta v}{2\sqrt{a_{\max} b_{\text{com}}}}. \tag{14}$$

Here, s_0 is the minimum spacing, Δv is the speed difference between the subject vehicle and its leader, and T is the desired time headway of the subject vehicle. To yield to a pedestrian, the subject vehicle treats the pedestrian as an obstacle and decelerates to stop. Other car following models can also be integrated into our system [11] [14].

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C. INTERACTION BETWEEN VEHICLES AND PEDESTRIANS

A vehicle makes decisions based on the perceived information from two kinds of regions: (1) the front region and (2) the adjacent regions. The closest agents in front of a vehicle will influence the speed of the vehicle. If there are no pedestrians in the front region of the vehicle, the motion of the vehicle is computed based on the car following model (see Section VI-B). Otherwise, pedestrians in the front of the vehicle will affect the speed of the vehicle when the vehicle is yielding or maintaining a safe distance from pedestrians. When pedestrians are in the vicinity of the vehicle, the vehicle assesses whether they are approaching the front area. Based on safety considerations, it then determines the appropriate action to take, such as slowing down or even decelerating to stop. Figure 4 illustrates some examples. Pedestrians at D_1 and C are positioned in front of vehicles 1 and 2, respectively, thereby influencing the behavior of the corresponding vehicles. Although no pedestrians are in front of vehicle 3, another pedestrian is walking toward location D_3. In this situation, the vehicle must stay aware to avoid a collision.

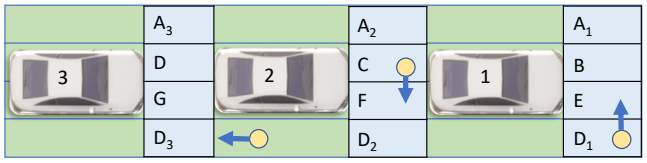


FIGURE 4. Perceived regions of a vehicle. A yellow disk represents a pedestrian and an arrow indicates the movement direction.

Figure 5 illustrates the interaction between a vehicle and a pedestrian. Denote d_{pet} and d_{veh} as the distances traveled by the pedestrian and the vehicle, respectively. d_{safe} is the comfortable safe distance between the pedestrian and the vehicle. The pedestrian moves with a speed $\mathbf{s}_{pet}$ to cross the road in front of a vehicle. Its temporary destination is $\mathbf{p}_{pet}$ which is a position at the layer above the vehicle. The pedestrian will travel a distance d_{pet} to $\mathbf{p}_{pet}$. $\mathbf{p}_{veh}$ is the future position of the vehicle. To avoid collision, the pedestrian should move to the safe position $\mathbf{p}_{pet}$ before the vehicle reaches position $\mathbf{p}_{veh}$.

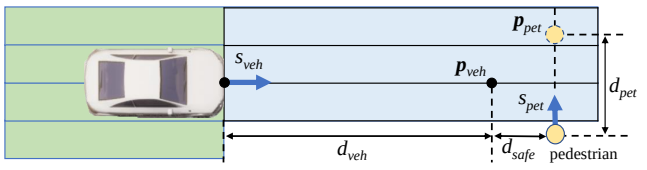


FIGURE 5. Vehicle-pedestrian interaction.

The vehicle can maintain its speed or yield to the pedestrian. To yield to the pedestrian, we adopt the car-following model to achieve. The times taken by the pedestrian and the vehicle to travel are $t_{pet} = d_{pet}/s_{max}$ and $t_{veh} = d_{veh}/s_{veh}$, respectively. Here, s_{max} is the maximum walking speed of the pedestrian. If $t_{veh} >= t_{pet}$, the vehicle yields to the pedestrian. A ghost point (i.e., invisible) with zero speed is placed at the middle point of the intersection region. The vehicle will gradually slow down to have a safe response. If $t_{veh} < t_{pet}$, the pedestrian has to wait. However, the vehicle may yield to the pedestrian based on a given probability distribution.

D. IMPLEMENTATION OF DYNAMIC GRAPH

At both ends of a road segment, we place ghost vehicles for each layer. Thus, we can always form a gap region for a vehicle. Each vehicle is assigned an identifier (ID). A dictionary is an unordered data structure for efficient data access. An element of a dictionary is a pair consisting of a unique key and a value. A value is a list in our case. We store the node set (V) and the edge set (E) as two dictionaries which are denoted as D_{node} and D_{edge}, respectively. A node stores the layer ID, and a pair of IDs of the leading and following vehicles. The IDs of the leading and following vehicles form the key of the corresponding node. Similarly, an edge connects two nodes and thus we use the IDs of the two nodes to form the key for the edge. The value associated with a node is the list of vehicles that form the gap region (i.e., the leader and the follower). And the value for an edge is the list of nodes that make up the edge. The dictionary D_{node} is updated during the simulation whenever any of the following three cases occur: (1) a vehicle changes layers; (2) a new vehicle enters the road; and (3) a vehicle moves outside of the scene. If a vehicle moves away from a layer, we remove all the nodes that are associated with the vehicle. Similarly, if a vehicle enters a layer, we determine the leading and following vehicles based on the relative longitudinal positions of the vehicles. After that, new nodes are created for the involved vehicles, and edges are updated accordingly. If vehicles do not change layers, the gap regions do not change too. However, the connectivity of the gap regions may be changed. So the edges and the dictionary D_{edge} are updated. Overall, the number of region gaps on a layer is directly proportional to the number of vehicles present on that layer.

VII. EXPERIMENTS AND RESULTS

We implemented the system on Carla simulator 0.9.12 with Python 3.8. The hardware configuration included an Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz, an NVIDIA GeForce RTX 2070 SUPER and 16.0 GB RAM. The default simulation time step was 0.1 second. The acceleration factor δ was set to 4 [64] for natural motion in our experiments (see Section VI-B). As there were five factors that could be set for simulating pedestrians' route decision making, the number of possible combinations was too large to report exhaustively. Therefore, we analyzed selected subsets to examine their influence on pedestrian behavior. We conducted a series of experiments to investigate key aspects of pedestrian behavior under varying conditions. The objectives of each experiment are summarized as follows:

Ex.1: Pedestrians' local route choice

Ex.2: Effects of simple factor combinations on pedestrian travel time and distance

Ex.3: Effects of various factor combinations on pedestrian travel time

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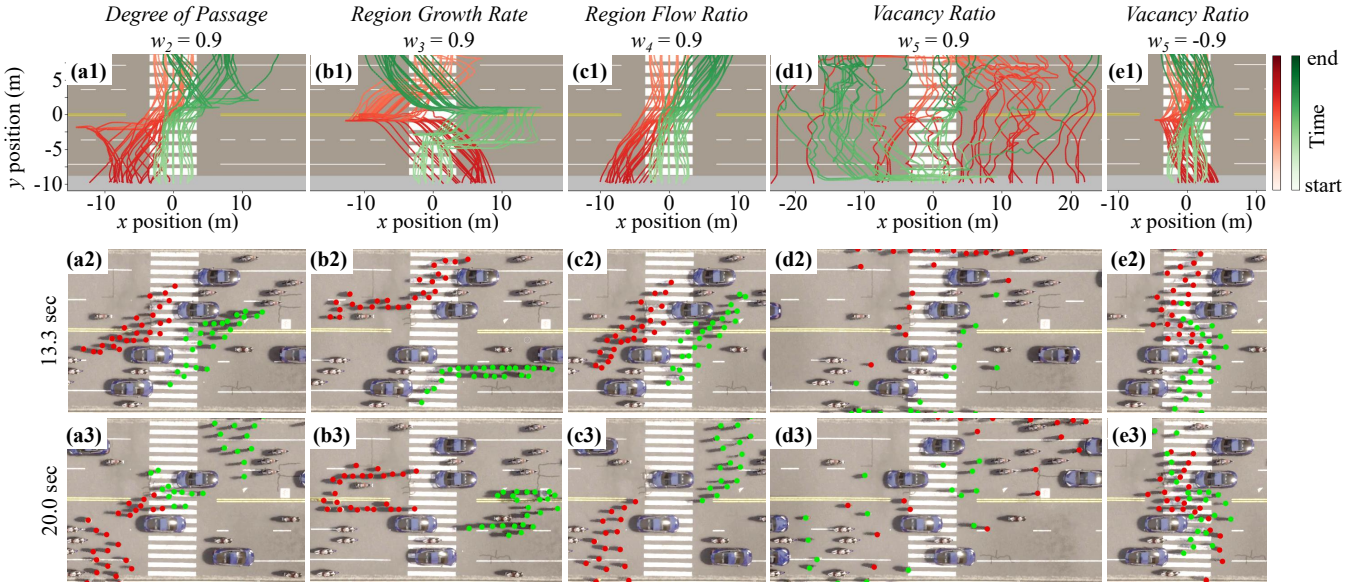


FIGURE 6. **Pedestrian trajectories. Pedestrians with red trajectories move down while pedestrians with green trajectories move up.**

Ex.4: Effects of pedestrian volume on pedestrian travel time
Ex.5: Vehicle deceleration when yielding pedestrians
Ex.6: Effects of pedestrian and vehicle arrival rates on delays
Ex.7: A three-phase crossing mechanism
Ex.8: System computational performance

Finally, we created animations for various applications and conducted a user-evaluation study to assess three of these animations. Hereafter, SD denotes the standard deviation.

A. EX.1: PEDESTRIANS' LOCAL ROUTE CHOICE

This experiment explored the trajectories of pedestrians crossing a road with conservative vehicles. *Adjusted Region Distance* was an important factor in pedestrians' decisions to cross the road. Pedestrians tended to select a gap region with a shorter travel distance compared to other gap regions. Additionally, for safety and travel cost reasons, a shorter path was preferred when two or more gap regions had the same preference level. We conducted a pairwise analysis to investigate the influence of the *Adjusted Region Distance* in combination with another factor on pedestrians' local route choices. This analysis was carried out under conditions where the contribution of *Adjusted Region Distance* was minimal (e.g., $w_1 = 0.1$), while the other factor played a dominant role (i.e., $w_j = 0.9$, where $j \in 2, 3, 4, 5$). Additionally, we examined pedestrian trajectories under a scenario where the *Vacancy Ratio* was assigned a negative weight, thereby promoting movement toward more crowded gap regions. These combinations isolated the effects of individual factors. Thus, we could clarify how each factor influenced pedestrian behavior independently. We simulated a scenario where there were 32 vehicles (12 cars and 20 motorcycles; Mean Speed = 5.6 m/s, SD = .2 m/s). The width of the road was 17.40 meters. There were 15 layers in the dynamic graph. On each side of the road, there were 25 pedestrians (Mean Speed = 1.2 m/s, SD = .2 m/s) [65]. We assumed some pedestrians did not obey the rules and might walk in front of vehicles [21].

Figures 6(a1-e1) show the trajectories of pedestrians. In addition, Figures 6(a2-e2) and (a3-e3) show the snapshots of the pedestrians represented as circles at two different simulation time stamps. When *Degree of Passage* (w_2) dominates, pedestrians choose a large gap region. The individual trajectories of pedestrians moving in the same direction are seen clearly (see Figure 6(a1)) as the pedestrians tend to move into spacious gap regions (see Figure 6(a2, a3)). When *Region Growth Rate* (w_3) dominates, the trajectories of the pedestrians along the longitudinal direction (i.e., along the vehicle movement direction) often appear (see Figure 6(b1)). This is because pedestrians are searching for expanding gap regions between vehicles. Pedestrians tend to move into gap regions where the oncoming vehicles are slowing down or the vehicles in front are accelerating (see Figure 6(b2, b3)). When *Region Flow Ratio* (w_4) dominates, pedestrians tend to move to the regions where more pedestrians move in the same direction or fewer pedestrians move in the opposite direction. Pedestrians moving in different directions are separated (see Figure 6(c1)) while pedestrians moving in the same direction are close to each other (see Figure 6(c2, c3)). The result shows that the pedestrians moved in a consistent bidirectional flow. When *Vacancy Ratio* (w_5) dominates, pedestrians tend to move to less occupied regions. Consequently, the pedestrians separated and scattered across the road, resulting in a poor traffic situation (see Figure 6(d1-d3)). However, when the *Vacancy Ratio* contributes negatively to the selection of gap regions, pedestrians tend to move in close proximity to one another (see Figure 6(e1–e3)). This scenario reflects behavior driven by herd mentality and safety concerns [52].

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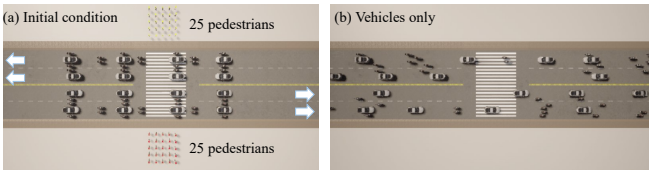


FIGURE 7. Scene model for studying effects on pedestrians.

B. EX.2: PEDESTRIANS' TRAVEL TIME AND TRAVEL DISTANCE

We studied how pedestrian travel time and travel distance were affected under two basic cases as well as various weights combinations in a traffic jam. The two basic cases gave a baseline for comparison. By understanding the outcomes under these two cases, we could better interpret the effects of more complex weight combinations. In addition, we investigated how various weight combinations affected pedestrian behavior. This allowed us to have a more comprehensive understanding of how different factors interacted. Non-linear effects and synergies of the factor combinations could be revealed.

In the scene, there was a four-lane road with vehicles traveling to the left in the top two lanes and to the right in the bottom two lanes (see Figure 7). The road segment was divided into 17 layers equally in the dynamic graph. Initially, there were 16 cars and 36 motorcycles. The length of the crosswalk was 20 meters. The mean vehicle speed was 5.6 m/s (SD = 0.2 m/s). There were 25 pedestrians at the top and bottom sides of the road, respectively. The destinations of the pedestrians were on the opposite side of the road. Pedestrians' walking speeds were uniformly sampled from the interval [1.1,1.2] m/s, with a standard deviation of 0.03 m/s. Thus, pedestrians with different speeds could have equal likelihood to perform certain behaviors. The simulation was finished when all the pedestrians reached the other side of the road.

Two Basic Cases: The first case was that the pedestrians selected regions based on *Adjusted Region Distance* only (**RD** condition). The second case was that all the factors had the same weight (**ALL** condition). In the **RD** condition, $w_1 = 1$ and the other four weights were zero. The average travel distance and travel time of the pedestrians were 25.41 m ($SD = 2.66$) and 23.15 s ($SD = 2.48$), respectively. In the **ALL** condition, we had $w_j = .2$ and $j \in \{1, 2, \cdots, 5\}$. The average travel distance and travel time were 26.71 m ($SD = 5.59$) and 24.84 sec ($SD = 5.93$), respectively. In the **RD** condition, the pedestrians' travel distance and time were shorter than in the **ALL** condition. This was because, in the **RD** scenario, pedestrians only moved towards the nearest region, and their waiting times were not too long. As a result, they experienced shorter travel distances and travel times compared to the **ALL** condition.

Effects of Two-Factor Combinations on Pedestrians: We studied the effects of *Adjusted Region Distance* and the other four factors on pedestrians. In this analysis, the weight of *Adjusted Region Distance* was varied as α, while the weight of another individual factor was set as $1 - \alpha$. By changing α systematically, we could explore how different levels of contribution of *Adjusted Region Distance* relative to another factor influence pedestrian behavior. Thus, we could identify the specific impact of *Adjusted Region Distance* in combination with other factors and observe the interplay between the factors. In addition to travel time and distance for pedestrians, we included search time, which is the amount of time a pedestrian spends moving within a layer while looking for a preferred gap region. This metric reflects hesitation during the crossing decision process.

Figures 8(a,b,c) shows the average travel time, travel distance, and search time for gap regions. In general, for a higher contribution of *Adjusted Region Distance*, the average travel time, travel distance, and search time for gap regions were lower than most of the other cases. When $w_1 = \alpha = .1$, the factors for achieving the lowest to longest average travel times were *Region Flow Ratio* (w_4), *Vacancy Ratio* (w_5), *Degree of Passage* (w_2), and *Region Growth Rate* (w_3). The fastest average travel time of pedestrians was about 20 seconds in the condition that $w_1 = \alpha = .9$ and $w_5 = .1$. In addition, the average travel distance and time spent in searching for gap regions were also the lowest. For a moderate or higher contribution of *Adjusted Region Distance* ($w_1 > .6$), the pedestrians would spend less travel time, travel distance, and time spent in searching for gap regions. For the effect of *Adjusted Region Distance* and *Region Flow Ratio*, the average travel time, travel distance, and search time for gaps were the lowest compared to the other factor combinations when the contribution of *Adjusted Region Distance* is lower ($w_1 < .6$) (see the red curves in Figures 8(a,b,c)). In general, the effects of different combinations of factors on the trends of pedestrian travel time, travel distance, and the time spent searching for gap regions were similar.

Summary of Impact on Travel Time and Distance: We summarize the primary influence of the five behavioral factors on pedestrian travel time and distance as follows.

I. Adjusted Region Distance: A higher *Adjusted Region Distance* may encourage more direct paths by reducing detours. However, if pedestrians move with greater caution, travel time may increase, especially as gap regions evolve dynamically.

II. Degree of Passage: A higher *Degree of Passage* affords easier navigation, typically supporting faster traversal. However, such regions may be located farther away, resulting in longer travel time.

III. Region Growth Rate: Rapidly expanding gap regions signal emerging crossing opportunities. Pedestrians need to respond quickly; otherwise, contraction of the gaps may lead to extended waiting times.

IV. Region Flow Ratio: A higher *Region Flow Ratio* indicates fewer conflicting movements, facilitating smoother crossings. However, congestion can still lead to prolonged travel times despite the favorable flow.

V. Vacancy Ratio: A higher *Vacancy Ratio* supports unobstructed paths, reducing stop-and-go movement and thereby

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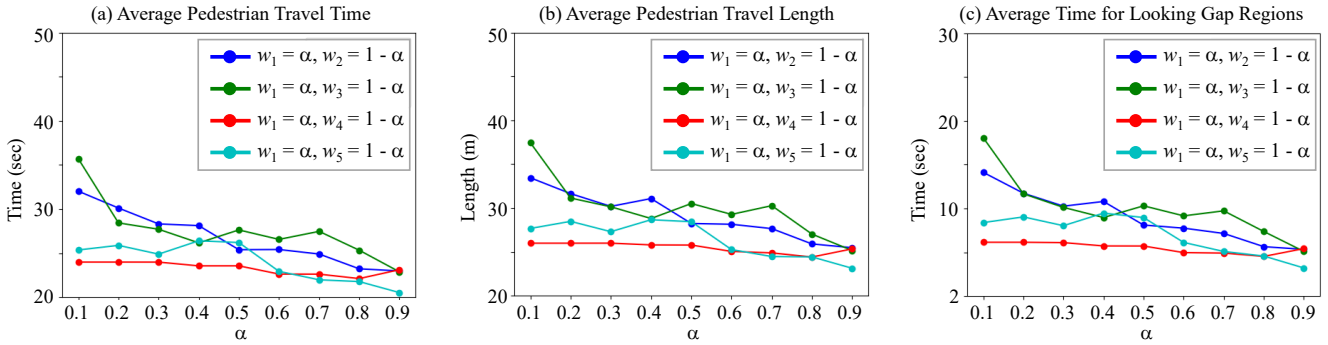


FIGURE 8. **Impact of *Adjusted Region Distance* and paired factors on pedestrians.**

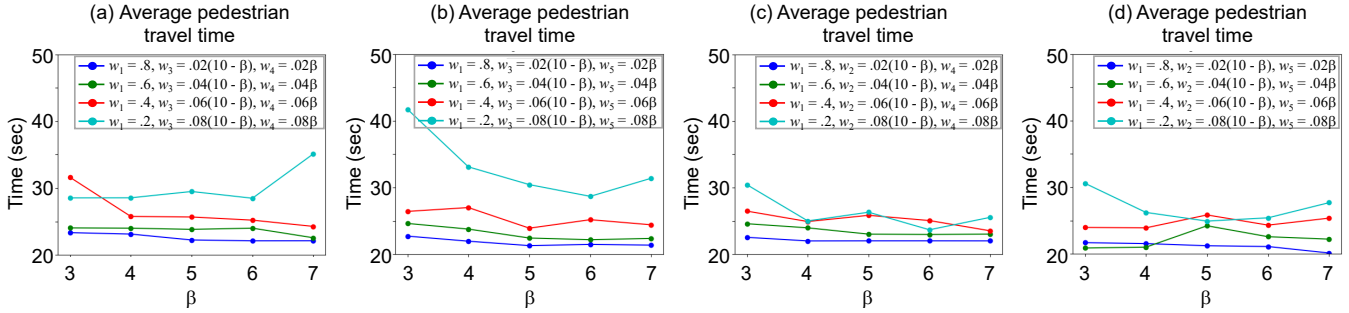


FIGURE 9. **Three-factor impact on pedestrian travel time.**

lowering travel time. However, travel distance varies depending on the proximity of viable gap regions. For example, gaps that are closer typically result in shorter routes.

C. EX.3: EFFECTS OF VARIOUS FACTOR COMBINATIONS ON PEDESTRIAN TRAVEL TIME

We investigated how the following factor pairs with *Adjusted Region Distance* affected the pedestrian travel time: {*Region Growth Rate* (w_3), *Region Flow Ratio* (w_4)}, {*Region Growth Rate* (w_3), *Vacancy Ratio* (w_5)}, {*Degree of Passage* (w_2), *Region Flow Ratio* (w_4)}, {*Degree of Passage* (w_2), *Vacancy Ratio* (w_5)}. The weight for *Adjusted Region Distance* was changed from .2 to .8. Figure 9 shows the average travel time. β was a parameter that was used to control the weights of the other two factors. For all the cases, the shortest average travel time was found (20 s) when pedestrians made decisions based mainly on *Adjusted Region Distance* ($w_1 = .8$), a moderate *Vacancy Ratio* ($w_5 = .14$), and a little *Degree of Passage* ($w_2 = .06$) (see Figure 9(d)). When $w_1 = .8$, the average travel time was deviated by about two seconds for each factor combination. The result showed that *Adjusted Region Distance* was a reliable factor for effective road crossing performance.

D. EX.4: EFFECT OF PEDESTRIAN VOLUME ON PEDESTRIAN TRAVEL TIME

We investigated how pedestrian volume affected the average pedestrian travel time for the number of pedestrians within [10, 50] under various factor combinations. Based on the results, we identified the interplay between different factors and their collective impact on pedestrian travel time.

In the scene, each side of the road had the same number of pedestrians. Figure 10 shows the average travel times under different combinations of the five factors. For most combinations, the impact on the average travel time was small when *Adjusted Region Distance* was dominant (e.g., $w_1 = .9$). Particularly, in the case that $w_5 = .1$ (*Vacancy Ratio*), the average travel time was almost not affected by the volume of the pedestrians. In these situations, the pedestrians moved to nearby gap regions while avoiding moving to congested gap regions. However, when the contribution of the *Adjusted Region Distance* was small, the average travel time was greatly affected. For example, when $w_1 = .1$ (*Adjusted Region Distance*) and $w_3 = .9$ (*Region Growth Rate*), the average travel time increased (from 23 s to 34 s) as the number of pedestrians increased. Hence, relying solely on the *Region Growth Rate* for decision-making showed unreliable due to its strong dependence on traffic flow.

An interesting situation occurred in the case that $w_1 = .1$ (*Adjusted Region Distance*) and $w_2 = .9$ (*Degree of Passage*). When the number of pedestrians increased to 50, the average travel time was lower than that when the number of pedestrians was 10. When *Degree of Passage* had a higher impact on pedestrians, the pedestrians tended to choose a larger gap region. The oncoming vehicles would slow down or stop when more pedestrians were crossing. Thus, the width of the gap regions increased while the leading vehicles were moving

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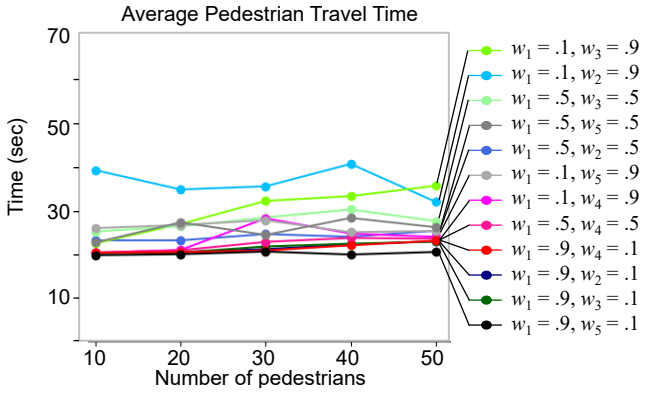


FIGURE 10. **Two-factor impact on pedestrian travel time.**

away, resulting in a higher *Degree of Passage*. The following pedestrians took less time to look for gap regions. Hence, the average travel time became shorter as more pedestrians were crossing.

E. EX.5: VEHICLE DECELERATION WHEN YIELDING PEDESTRIANS

When pedestrians crossed a road in front of vehicles, the vehicles needed to decelerate to keep a safe distance from the pedestrian to avoid accidents. We recorded the maximum deceleration of vehicles traveling within three different desired speed intervals: [1, 2], [3, 4], and [5, 6] m/s. Table 1 shows the results for the combinations of three factors which were *Adjusted Region Distance*(w_1), *Region Growth Rate*(w_3), and *Region Flow Ratio*(w_4) with different weights. The result shows that when the vehicles were moving at a higher speed (i.e., [5,6] m/s), a greater deceleration was required to yield to the pedestrians compared to those with a lower speed (i.e., [1,4] m/s). The result underscores the importance of vehicle awareness and caution in mid-block areas. Additionally, diverse visual animations could be created for these vehicle-pedestrian interactions.

TABLE 1. **Vehicle maximum deceleration.**

Factor Combination	Vehicle speed interval (m/s)		
	[1,2]	[3,4]	[5,6]
	Maximum deceleration (m/s^2)		
$w_1 = .2, w_3 = .24, w_4 = .56$	1.63	2.95	**4.28**
$w_1 = .2, w_3 = .56, w_4 = .24$	1.63	2.86	**5.05**
$w_1 = .4, w_3 = .18, w_4 = .42$	1.64	2.91	**3.89**
$w_1 = .4, w_3 = .42, w_4 = .18$	1.74	2.93	**4.11**
$w_1 = .6, w_3 = .12, w_4 = .28$	1.64	2.91	**3.87**
$w_1 = .6, w_3 = .28, w_4 = .12$	1.64	2.91	**3.89**
$w_1 = .8, w_3 = .06, w_4 = .14$	1.64	2.91	**3.87**
$w_1 = .8, w_3 = .14, w_4 = .06$	1.58	2.97	**3.98**

F. EX.6: EFFECTS OF PEDESTRIAN AND VEHICLE ARRIVAL RATES ON DELAYS

The frequency with which agents appear is called the arrival rate of agents. We investigated how the arrival rates of the pedestrians and vehicles affect on the delay times of pedestrians and vehicles. Denote the arrival rates of pedestrians and vehicles as λ_{ped} and λ_{vec}, respectively. The desired vehicle speeds were uniformly sampled from the interval $[3,4]$ m/s. Denote t_1^p and t_2^p as the average travel times of the pedestrians in the presence and absence of vehicles, respectively. Similarly, denote t_1^v and t_2^v as the average travel times of the vehicles in the presence and absence of pedestrians, respectively. The average delay times of the pedestrians and vehicles were computed as $\Delta t^p = t_1^p - t_2^p$ and $\Delta t^v = t_1^p - t_2^p$, respectively. Figure 11 presents the effects of varying pedestrian and vehicle arrival rates on the delay times encountered by both pedestrians and vehicles. The result was obtained for 300 s in simulation. For a higher pedestrian arrival rate, the average delay time of pedestrians increased slightly (see Figure 11(a)) because of congestion. When a pedestrian started to cross, other pedestrians could follow the leader pedestrian. Thus, the average waiting time of pedestrians was reduced. However, the average delay time increased when the vehicle arrival rate increased (see Figure 11(b)). This was because when the number of vehicles increased, the chance of vehicles blocking pedestrians increased and pedestrians had to spend more time waiting for available gaps. As the pedestrian volume increased, the vehicles spent a longer time waiting for the pedestrians. Thus, the average delay time of the vehicles increased (see Figure 11(c)). The results showed similar effects on pedestrian and vehicle delay times as found in previous studies [52] [32].

G. EX.7: A THREE-PHASE CROSSING MECHANISM

While pedestrians cross a road, the entire process can be divided into three phases [21], which are approaching, appraising, and crossing. In the approaching phase, the pedestrians move towards the crosswalk on the sidewalk at a relatively steady speed. In the appraising phase, they slow down to approach the curb to assess whether it is safe to cross. Finally, in the crossing phase, they increase their speed to cross the road. The simulation setting for the pedestrian walking speed was as follows: approaching (Mean = 1.28 m/s, SD = .18 m/s), appraising (Mean = .86 m/s, SD = .10 m/s), and crossing (Mean = 1.45 m/s, SD = .18 m/s). In the scene, two roads were separated by a refuge island (width = 3.70 m). The distance between the two sides of the two roads was 20.5 m. There were 13 layers in the dynamic graph. 17 pedestrians were crossing on both sides of the road. 5 vehicles and 3 motorcycles were simulated. We selected three combinations of factors to study pedestrian crossing behavior: (I) {*Adjusted Region Distance* ($w_1 = 1$)}, (II) {*Adjusted Region Distance* ($w_1 = 0.3$), *Region Growth Rate* ($w_3 = 0.7$)}, and (III) {*Adjusted Region Distance* ($w_1 = 0.3$), *Region Flow Ratio* ($w_4 = 0.7$)}. These combinations represented the same level of influence from *Region Growth Rate* and *Region Flow Ratio*, with a consistent moderate contribution from *Adjusted Region Distance*. They represented three distinct strategies: (I) A proximity-based strategy, where pedestrians naively choose to cross based solely on the shortest distance

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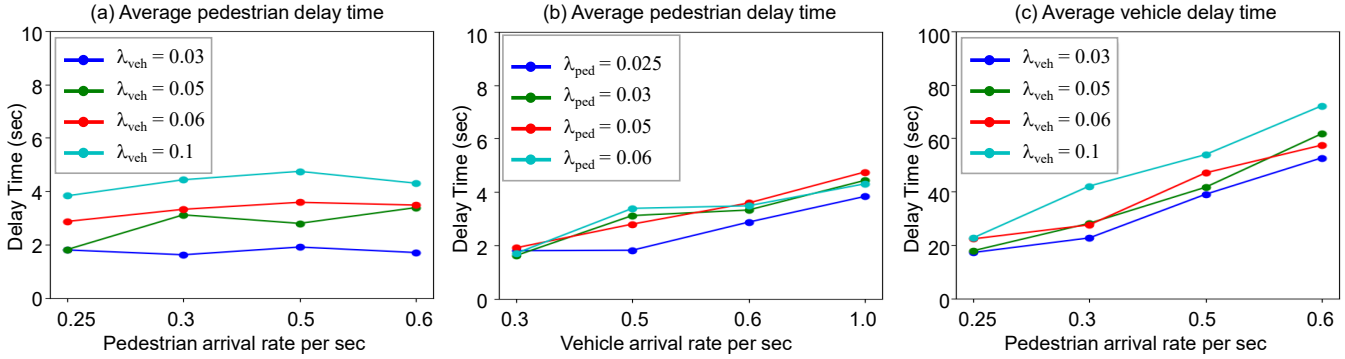


FIGURE 11. **Impact of pedestrian and vehicle arrival rates on delay time.**

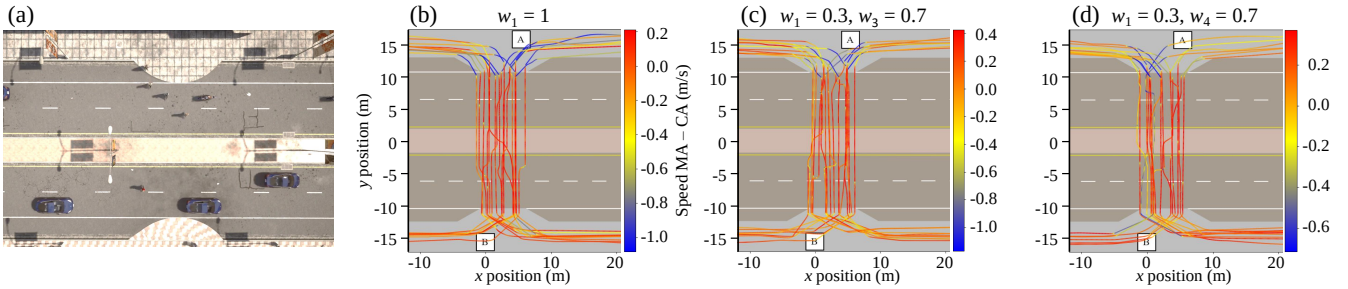


FIGURE 12. **Trajectories of pedestrians in three conditions. The pedestrians move from the location A to the location B.**

to a gap region; (II) An anticipatory strategy, which selects nearby gap regions while also considering their growth rate, reflecting pedestrians' predictive behavior to avoid narrowing spaces; and (III) A flow-aware strategy, which favors closer gap regions that are less obstructed, indicating sensitivity to dynamic flow conditions.

Denote CA and MA for a pedestrian as the average pedestrian speed in the entire simulation (or cumulative average speed) and the average speed of three frames (or moving average speed), respectively. We used both CA and MA to investigate pedestrian speed dynamics. CA showed long-term trends while MA captured immediate changes. Comparing CA and MA revealed speed fluctuations, highlighting acceleration and deceleration periods. We could observe how pedestrians adjusted their speeds in response to dynamic interactions with vehicles and pedestrians. Figure 12 shows the results for pedestrians. The color-coded trajectories show the speed difference of the pedestrians individually between MA and CA. A reddish part indicates movement at a higher speed than the average, whereas a bluish or yellowish part indicates movement at a lower speed than the average. As can be seen, as pedestrians moved towards the curb on the sidewalk, their speed remained relatively steady. When they were close to the curb, they slowed down significantly to carefully evaluate the road condition. Then they decided to cross the road and speed up to avoid staying on the road. After they reached the opposite side of the road, they slowed down in a safe area. Some pedestrians maintained a high walking speed to reach their destination quickly. Our results clearly showed the speed differences of the pedestrians across the three stages of crossing, aligning with previous findings [21].

H. EX.8: SYSTEM COMPUTATIONAL PERFORMANCE

We evaluated the performance of our system for the simulation process in terms of the frame rate per second (fps). The dimension of the scene was 40m x 80m. The number of pedestrians was set between 20 and 100. If a vehicle or pedestrian moved out of the scene, a new vehicle or pedestrian was simulated. In this way, the number of pedestrians and vehicles was maintained in the scene. During the simulation process, we disabled the rendering computation. Figure 13 shows the result. The system achieved 30 frames per second if the number of pedestrians was fewer than 80. A higher frame rate was achieved by simulating fewer pedestrians. However, the system was implemented in Python which is a scripting language. Thus, for efficient computation, other programming languages such as C/C++ can be adopted.

I. APPLICATIONS

We employed the factor-based approach to simulate four scenarios which were *Daily Pedestrian Crossings* (40 m x 80 m), *Pedestrian Challenges* (30 m x 70 m), *Emergency Pedestrian Rush* (40 m x 80 m), and *User Control* (70 m x 70 m) (see Figure 14). In *Daily Pedestrian Crossings*, the pedestrian crossed a road with four lanes. In the upper two lanes, the vehicles moved from right to left. In the lower two lanes, vehicles moved from left to right. Some pedestrians choose to cross the road directly to save time. (Mean Speed

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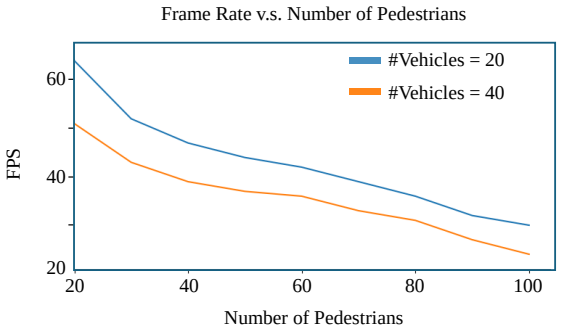


FIGURE 13. Simulation frame rates.

= 1.5 m/s, SD = .15 m/s). Nevertheless, they crossed the road with confidence, assuming that the vehicles would give way to them. In *Pedestrian Challenges*, some vehicles were parked on the curb and two vehicles stopped in two lanes. Some pedestrians quickly move to the other side to avoid staying on the roadway for too long, while vehicles change lanes to bypass the blocked vehicles. (Mean Speed 1.8 m/s, SD = .15 m/s). In *Emergency Pedestrian Rush*, an emergency occurred in the buildings at the upper part of the scene. The pedestrians escaped away from the buildings and some crossed the road (Mean Speed = 3.5 m/s, SD = .08 m/s). The vehicles were moving slowly to allow the pedestrians to reach the other side of the road safely. In *User Control*, a pedestrian was controlled by a user to cross a curved road with four lanes. The oncoming vehicles reacted appropriately to avoid a collision with the pedestrian. In all the scenarios, the movement speeds of pedestrians and vehicles were tunable to achieve desired results.

J. USER-EVALUATION STUDY

We conducted a user-evaluation study to evaluate our system. We recruited 17 participants (with labels P1 to P17), including 13 males and 4 females, with ages ranging from 23 to 58 (Mean = 28.7, SD = 8.5). Three participants (P3, P5, P14) had prior experience in animation production or related fields, while others had no such experience. Furthermore, three participants (P4, P10, P11) had no driving experience, while the rest had experience driving motorcycles or sedans. The participants evaluated three animations (with labels S1 to S3) one by one, including *Daily Pedestrian Crossings*, *Pedestrian Challenges*, and *Emergency Pedestrian Rush* (see Figure 14(S1,S2,S3)). In *Daily Pedestrian Crossings*, the pedestrians cross a road during non-peak hours without designated crossings. In *Pedestrian Challenges*, vehicles encounter road obstructions caused by stalled vehicles. However, pedestrians may cross unpredictably. Thus, the vehicles maneuver around the obstacles while adjusting their speeds to avoid colliding with the pedestrians. In *Emergency Pedestrian Rush*, pedestrians rush onto the road during an emergency. For each animation, the participants followed the following procedure for evaluation:

TABLE 2. Questionnaire of the user-evaluation study.

No	Questions
Q1	Do you agree that the animation of the pedestrians accurately reflects the provided background story?
Q2	Do you agree that the animation of the vehicles accurately reflects the provided background story?
Q3	Do you agree that the interactions between pedestrians and vehicles are natural?
Q4	How would you characterize the behavior of pedestrians?
Q5	How would you characterize the behavior of vehicles?

TABLE 3. Mean scores for the three animations.

	S1		S2		S3	
No	Scores	SD	Scores	SD	Scores	SD
Q1	6.2	0.7	6.2	0.9	6.1	1.1
Q2	6.4	0.6	6.1	1.0	6.1	0.9
Q3	5.6	1.0	5.5	0.9	5.1	1.0
Q4	5.6	1.0	5.0	0.9	6.3	1.3
Q5	3.2	1.8	2.9	1.4	3.4	1.4

1) Overview (60 s). Read the background story for the animation.
2) Watching (60 s). Watch the animation.
3) Questionnaire (180 s). Complete a 5-question survey using a 7-point Likert scale. Questions 1, 2, and 3 assess the coherence and naturalness of the animation (1 = strongly disagree, 4 = neutral, 7 = strongly agree). Questions 4 and 5 evaluate the aggressiveness of pedestrian and vehicle behaviors (1 = very conservative, 4 = normal, 7 = very aggressive). Table 2 shows the questionnaire.

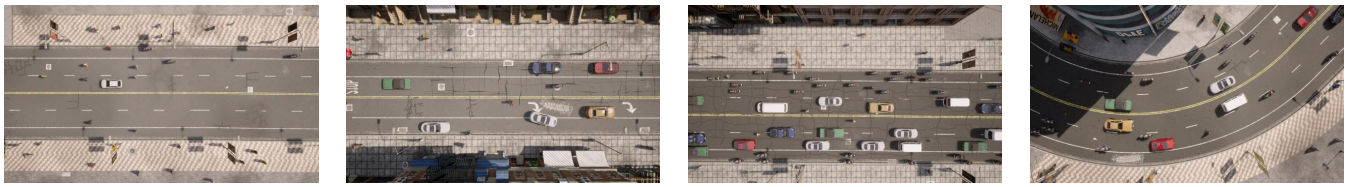
Table 3 shows the result. For **Q1** and **Q2**, the mean scores were consistently high (ranging from 6.1 to 6.4), demonstrating strong agreement that the animations accurately reflected the background story for both pedestrians and vehicles. For Q3, the scores were slightly lower in the *S3* (5.1) compared to *S1* (5.6) and *S2* (5.5), indicating that the interaction in dynamic and complex environment presented more challenges. For **Q4**, the pedestrian behavior became progressively more aggressive, with scores increasing from 5.0 in *S2* to 6.3 in *S3*, aligning with the expected urgency in the *Emergency Pedestrian Rush* scenario. In contrast, vehicle behavior (**Q5**) remained cautious across all the scenarios (Mean Score ranging from 2.9 and 3.4), reflecting the safe driving practices.

K. DISCUSSION AND LIMITATIONS

In this study, we focus on simulating pedestrians' local route choice decisions by proposing five key behavioral factors based on the characteristics of gap regions. When multiple gap regions exist on a layer, we compute an aggregate value, termed the *Degree of Preference*, based on the contributions of these five factors. Our main goal is not to match real-world data. Instead, we aim to provide animators with flexibility to adjust these factors to achieve desirable animation results. The five factors are intuitive and visually interpretable in traffic animation. Animators can create diverse pedestrian crossing behaviors. These behaviors can range from conser-

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S1: Daily pedestrian crossings S2: Pedestrian challenges S3: Emergency pedestrian rush S4: User control

FIGURE 14. Applications scenarios.

vative, group-dependent, hesitant, individualistic, risk-taking, short-cut, socially aware, emergency-oriented.

In general, pedestrians will choose the closest gap region to cross, provided that the gap region is large enough for them to pass (*Degree of Passage*). However, if the *Vacancy Ratio* of a gap region is too low, they may not be able to move into the gap region. Of course, they can follow others to move together if the *Region Flow Ratio* aligns with their movement directions. In addition, if a gap region has a higher *Region Growth Rate*, the gap region will become larger to accommodate more pedestrians. Specifically, pedestrians with different behavioral tendencies influence road crossing decisions. Individualistic pedestrians prefer walking alone and thus favor gap regions with higher *Vacancy Ratios*. In contrast, group-dependent pedestrians seek the presence of others and are more likely to choose regions with lower *Vacancy Ratios*. Risk-taking pedestrians may opt for gap regions with lower *Region Passage* and tend to exploit changing *Region Growth Rate*. Shortcut-seeking pedestrians prioritize proximity (i.e., *Adjusted Region Distance*) and are willing to blend into traffic. Socially aware pedestrians aim to maintain personal space (i.e., moderate *Vacancy Ratio*) and avoid obstructing others (i.e., moderate *Region Flow Ratio*). Finally, emergency-oriented pedestrians seize any available opportunity to cross, regardless of conditions. In summary, these five factors influence pedestrian route decisions. The combinations of these factors ensure dynamic and adaptable crossing behavior of the pedestrians.

This study focuses on a local route plan for pedestrians. If vehicles do not move, a long-term route plan for pedestrians with the least effort may be possible. However, if the vehicles move, there will be uncertainty because of changes in the characteristics of gap regions and the regions' connectivity. A long-term plan is difficult to make. Thus, we focus on handling local route decisions for pedestrians in a moderately changing traffic flow.

Our method has the following limitations. The proposed approach can accurately track gap regions formed by leaders and their following vehicles along layers of road segments. However, several problems may occur because of tracking vehicles along these discrete layers. First, for a layer with a large width, when a vehicle overlaps with the layer, the associated overlapped region is marked as not movable. Thus, pedestrians cannot walk the gap region overlapped by the vehicle. To accurately capture the movable gap regions of pedestrians along a layer, a smaller width for a layer is required but computation cost may be increased. Second, we adopted the combinations of the five factors to evaluate the degrees of preference for gap regions. However, this does not imply that the factors contribute equally to affect pedestrians' behaviors when the factors have the equal weights. This is because some factors may be more visually or contextually dominant than others. For example, pedestrians can see how far they can reach a region gap based on the *Adjusted Region Distance*. On the other hand, pedestrians can follow a crowd to move together based on *Region Flow Ratio* of a gap region even though the passage of the gap region looks small. Third, the focus of the current study is on visual plausibility rather than precise traffic simulation accuracy. Furthermore, pedestrian behavior varies across cultural and geographic contexts. Future research could investigate how this approach might be adapted or calibrated to reflect these variations, enabling realistic simulations for traffic analysis and urban planning. Finally, pedestrian behavior is influenced not only by the characteristics of gap regions, but also by other factors, such as individual personality traits, emotional states, surrounding environmental conditions, and the types of interacting vehicles. Future work can incorporate these influences to enable an intuitive design of pedestrian crossing behavior.

VIII. CONCLUSION AND FUTURE WORK

We have presented a factor-based approach for generating diverse interaction behaviors between pedestrians and vehicles in non-signalized mid-block regions, covering both conservative and risky interactions. The main idea is that a dynamic graphic enables pedestrians to perform local route decision-making in mixed traffic. The dynamic graph maintains the gap regions formed by leader and follower vehicles. Pedestrians make crossing decisions based on five intuitive factors characterizing the gap regions: adjusted region distance, passage ratio, region flow rate, region growth rate, and vacancy ratio. By specifying the desired influence of these factors, pedestrians adopt corresponding crossing strategies that reflect their behavioral preferences. The experiment results showed that the adjusted region distance was one of the most important factors that affected the stability of pedestrians' decision-making. A variety of pedestrian behaviors could be generated, such as moving together, moving away from congested regions, and wait-and-go behaviors. Our method has potential applications in generating datasets for training self-driving

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systems [66] that require diverse pedestrian interaction behaviors, including common, risky, and rare scenarios [67]. However, its applicability to real-world autonomous driving contexts remains to be validated through further investigation.

For future work, we would like to extend our approach to designing pedestrian-vehicle interaction patterns to achieve desirable visual effects in a variety of conditions, such as congested and hazardous scenarios. Additionally, we plan to incorporate dynamic graphs into a traffic authoring system to enable the creation and editing of customizable traffic scenarios. Large Language Models (LLMs) can perform a wide range of tasks in response to natural language prompts [68], enabling intuitive text-based interaction. Building on this capability, we propose extending our system to accept natural language input. The LLMs would analyze the input and extract relevant pedestrian-vehicle interaction details. Then the system could automatically generate corresponding animations based on the analysis. Finally, we aim to integrate crowd simulation techniques [69] and emotion contagion models [70] to simulate crowd evacuations in traffic scenarios involving diverse vehicle types and dynamic behaviors. These scenarios include complex emergencies such as vehicular accidents, explosions, terrorist attacks, and fires. Our system could enable the investigation of how different pedestrian behaviors influence evacuation efficiency. Potential applications span crowd evacuation planning, education, training, and immersive entertainment.

REFERENCES

[1] V Shreyas, Skanda N Bharadwaj, S Srinidhi, KU Ankith, and AB Rajendra. Self-driving cars: An overview of various autonomous driving systems. *Advances in Data and Information Sciences*, pages 361–371, 2020.

[2] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. Carla: An open urban driving simulator. In *Conference on robot learning*, pages 1–16. PMLR, 2017.

[3] Mahmud Hasan, Daniel Perez, Yuzhong Shen, and Hong Yang. Distributed microscopic traffic simulation with human-in-the-loop enabled by virtual reality technologies. *Advances in Engineering Software*, 154:102985, 2021.

[4] Shane Brown, Chelsea Nicholas, and Michael Kyte. Evaluating the effectiveness of dynamic traffic animations: Case study in transportation engineering education. *Journal of Professional Issues in Engineering Education and Practice*, 139(3):196–205, 2013.

[5] Guannan Lou, Yao Deng, Xi Zheng, Mengshi Zhang, and Tianyi Zhang. Testing of autonomous driving systems: where are we and where should we go? In *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pages 31–43, 2022.

[6] Tantan Zhang, Haipeng Liu, Weijie Wang, and Xinwei Wang. Virtual tools for testing autonomous driving: A survey and benchmark of simulators, datasets, and competitions. *Electronics*, 13(17):3486, 2024.

[7] Nirajan Shiwakoti, Richard Tay, and Peter Stasinopoulos. Development, testing, and evaluation of road safety poster to reduce jaywalking behavior at intersections. *Cognition, Technology & Work*, 22:389–397, 2020.

[8] Qianwen Chao, Huikun Bi, Weizi Li, Tianlu Mao, Zhaoqi Wang, Ming C Lin, and Zhigang Deng. A survey on visual traffic simulation: Models, evaluations, and applications in autonomous driving. In *Computer Graphics Forum*, volume 39, pages 287–308. Wiley Online Library, 2020.

[9] Mark Brackstone and Mike McDonald. Car-following: a historical review. *Transportation Research Part F: Traffic Psychology and Behaviour*, 2(4):181–196, 1999.

[10] Martin Treiber, Ansgar Hennecke, and Dirk Helbing. Congested traffic states in empirical observations and microscopic simulations. *Physical review E*, 62(2):1805, 2000.

[11] Arne Kesting, Martin Treiber, and Dirk Helbing. Enhanced intelligent driver model to access the impact of driving strategies on traffic capacity. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 368(1928):4585–4605, 2010.

[12] Jingjing Shen and Xiaogang Jin. Detailed traffic animation for urban road networks. *Graphical Models*, 74(5):265–282, 2012.

[13] Wen-Chieh Lin, Sai-Keung Wong, Cheng-Hsing Li, and Richard Tseng. Generating believable mixed-traffic animation. *IEEE Transactions on Intelligent Transportation Systems*, 17(11):3171–3183, 2016.

[14] Wen-Xing Zhu and H Michael Zhang. Analysis of mixed traffic flow with human-driving and autonomous cars based on car-following model. *Physica A: Statistical Mechanics and its Applications*, 496:274–285, 2018.

[15] Sean Curtis, Basim Zafar, Adnan Gutub, and Dinesh Manocha. Right of way. *The Visual Computer*, 29(12):1277–1292, 2013.

[16] Yan Mao, Zuning Li, Yongjian Li, and Wu He. Emotion-based diversity crowd behavior simulation in public emergency. *The Visual Computer*, 35(12):1725–1739, 2019.

[17] Tarik Hadzic, Kenneth N Brown, and Cormac J Sreenan. Real-time pedestrian evacuation planning during emergency. In *IEEE International Conference on Tools with Artificial Intelligence*, pages 597–604, 2011.

[18] Sai-Keung Wong, Yu-Shuen Wang, Pao-Kun Tang, and Tsung-Yu Tsai. Optimized evacuation route based on crowd simulation. *Computational Visual Media*, 3(3):243–261, 2017.

[19] Yi-Shin Lin, Aravinda Ramakrishnan Srinivasan, Matteo Leonetti, Jac Billington, and Gustav Markkula. A utility maximization model of pedestrian and driver interactions. *IEEE Access*, 10:118888–118899, 2022.

[20] Theodore A Petritsch, Bruce W Landis, Peyton S McLeod, Herman F Huang, Srikalyan Challa, and Martin Guttenplan. Level-of-service model for pedestrians at signalized intersections. *Transportation research record*, 1939(1):54–62, 2005.

[21] Andrea Gorrini, Luca Crociani, Giuseppe Vizzari, and Stefania Bandini. Observation results on pedestrian-vehicle interactions at non-signalized intersections towards simulation. *Transportation research part F: traffic psychology and behaviour*, 59:269–285, 2018.

[22] Akash Jain, Ankit Gupta, and Rajat Rastogi. Pedestrian crossing behaviour analysis at intersections. *International Journal for Traffic and Transport Engineering*, 4(1):103–116, 2014.

[23] Anye Zhou, Yongyang Liu, Einat Tenenboim, Shubham Agrawal, and Srinivas Peeta. Car-following behavior of human-driven vehicles in mixed-flow traffic: A driving simulator study. *IEEE Transactions on Intelligent Vehicles*, 8(4):2661–2673, 2023.

[24] Victor Cantillo, Julian Arellana, and Manuel Rolong. Modelling pedestrian crossing behaviour in urban roads: A latent variable approach. *Transportation research part F: traffic psychology and behaviour*, 32:56–67, 2015.

[25] Yanxiang Ye. A review of pedestrian crossing research based on traffic behavior. *Scientific Journal of Intelligent Systems Research Volume*, 4(6), 2022.

[26] Avinash Chaudhari, Shriniwas Arkatkar, Gaurang Joshi, and Manoranjan Parida. Exploring stage-wise pedestrian-crossing behavioral patterns at vulnerable urban midblocks: A perspective under heterogeneous traffic conditions. *Journal of Transportation Safety & Security*, 12(7):863–891, 2020.

[27] Gwenola Thomas and Stéphane Donikian. Virtual humans animation in informed urban environments. In *Proceedings Computer Animation 2000*, pages 112–119. IEEE, 2000.

[28] Michiel M. Minderhoud and Piet H. L. Bovy. Extended time-to-collision measures for road traffic safety assessment. *Accident Analysis & Prevention*, 33(1):89–97, January 2001.

[29] Hui Wang, Hongjia Zhang, Menglu Gu, and Chang Wang. Comparison of the pedestrian crossing intention parameters and research on intention recognition model under different road conditions. *IEEE Access*, 8:196675–196689, 2020.

[30] George Yannis, Eleonora Papadimitriou, and Athanasios Theofilatos. Pedestrian gap acceptance for mid-block street crossing. *Transportation planning and technology*, 36(5):450–462, 2013.

[31] Qianwen Chao, Zhigang Deng, and Xiaogang Jin. Vehicle–pedestrian interaction for mixed traffic simulation. *Computer Animation and Virtual Worlds*, 26(3-4):405–412, 2015.

[32] Peng Chen, Chaozhong Wu, and Shunying Zhu. Interaction between vehicles and pedestrians at uncontrolled mid-block crosswalks. *Safety science*, 82:68–76, 2016.

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[33] Lili Lu, Gang Ren, Wei Wang, Ching-Yao Chan, and Jian Wang. A cellular automaton simulation model for pedestrian and vehicle interaction behaviors at unsignalized mid-block crosswalks. *Accident Analysis & Prevention*, 95:425–437, 2016.

[34] Jing Zhao, Jairus Odawa Malenje, Jingxian Wu, and Ruoming Ma. Modeling the interaction between vehicle yielding and pedestrian crossing behavior at unsignalized midblock crosswalks. *Transportation research part F: traffic psychology and behaviour*, 73:222–235, 2020.

[35] Dongfang Yang, Ümit Özgüner, and Keith Redmill. Social force based microscopic modeling of vehicle-crowd interaction. In *2018 IEEE Intelligent Vehicles Symposium (IV)*, pages 1537–1542. IEEE, 2018.

[36] Qianwen Chao, Xiaogang Jin, Hen-Wei Huang, Shaohui Foong, Lap-Fai Yu, and Sai-Kit Yeung. Force-based heterogeneous traffic simulation for autonomous vehicle testing. In *2019 International Conference on Robotics and Automation (ICRA)*, pages 8298–8304. IEEE, 2019.

[37] Dongfang Yang, Ümit Özgüner, and Keith Redmill. A social force based pedestrian motion model considering multi-pedestrian interaction with a vehicle. *ACM Transactions on Spatial Algorithms and Systems (TSAS)*, 6(2):1–27, 2020.

[38] Pei-Yang Wu, Ren-Yong Guo, Liang Ma, Bin Chen, Junjie Wu, and Qiuhong Zhao. Simulation of pedestrian route choice with local view: A potential field approach. *Applied Mathematical Modelling*, 92:687–709, 2021.

[39] Shashikant N Sharma and Dungar Singh. Understanding mid-block traffic analysis: A crucial tool for road safety. *Think India Journal*, 26(3):5–9, 2023.

[40] Saad AH AlGadhi and Hani S Mahmassani. Simulation of crowd behavior and movement: Fundamental relations and application. *Transportation Research Record*, (1320), 1991.

[41] VJ Blue, MJ Embrechts, and JL Adler. Cellular automata modeling of pedestrian movements. In *1997 IEEE International Conference on Systems, Man, and Cybernetics. Computational Cybernetics and Simulation*, volume 3, pages 2320–2323. Ieee, 1997.

[42] Dirk Helbing and Peter Molnar. Social force model for pedestrian dynamics. *Physical review E*, 51(5):4282, 1995.

[43] Mehdi Moussaïd, Dirk Helbing, Simon Garnier, Anders Johansson, Maud Combe, and Guy Theraulaz. Experimental study of the behavioural mechanisms underlying self-organization in human crowds. *Proceedings of the Royal Society B: Biological Sciences*, 276(1668):2755–2762, 2009.

[44] Jooyong Lee, Taewan Kim, Jin-Hyuk Chung, and Jinho Kim. Modeling lane formation in pedestrian counter flow and its effect on capacity. *KSCE Journal of Civil Engineering*, 20(3):1099–1108, 2016.

[45] Zhilu Yuan, Hongfei Jia, Linfeng Zhang, and Lei Bian. Simulation of pedestrian behavior in the collision-avoidance process considering their moving preferences. *Discrete Dynamics in Nature and Society*, 2017(1):3678268, 2017.

[46] Kayvan Aghabayk, Majid Sarvi, and William Young. A state-of-the-art review of car-following models with particular considerations of heavy vehicles. *Transport reviews*, 35(1):82–105, 2015.

[47] Peter G Gipps. A behavioural car-following model for computer simulation. *Transportation Research Part B: Methodological*, 15(2):105–111, 1981.

[48] Luigi Pariota and Gennaro Nicola Bifulco. Experimental evidence supporting simpler action point paradigms for car-following. *Transportation research part F: traffic psychology and behaviour*, 35:1–15, 2015.

[49] Haiming Hao, Wanjing Ma, and Hongfeng Xu. A fuzzy logic-based multiagent car-following model. *Transportation Research Part C: Emerging Technologies*, 69:477–496, 2016.

[50] WenJing Wu, RunChao Chen, Hongfei Jia, Yongxing Li, and ZhiKang Liang. Game theory modeling for vehicle–pedestrian interactions and simulation based on cellular automata. *International Journal of Modern Physics C*, 30(04):1950025, 2019.

[51] Xiuying Xin, Ning Jia, Liang Zheng, and Shoufeng Ma. Power-law in pedestrian crossing flow under the interference of vehicles at an unsignalized midblock crosswalk. *Physica A: Statistical Mechanics and its Applications*, 406:287–297, 2014.

[52] Sheng Jin, Xiaobo Qu, Cheng Xu, and Dian-Hai Wang. Dynamic characteristics of traffic flow with consideration of pedestrians road-crossing behavior. *Physica A: Statistical Mechanics and its Applications*, 392(18):3881–3890, 2013.

[53] Yi Han, Qianwen Chao, and Xiaogang Jin. A simplified force model for mixed traffic simulation. *Computer Animation and Virtual Worlds*, 32(1):e1974, 2021.

[54] Yan Zhang, Xingguo Zhang, Yohei Fujinami, and Pongsathorn Raksincharoensak. Social force model-based adaptive parameters collision avoidance method considering motion uncertainty of the pedestrian. *IEEE Access*, 2023.

[55] Weiliang Zeng, Peng Chen, Hideki Nakamura, and Miho Iryo-Asano. Application of social force model to pedestrian behavior analysis at signalized crosswalk. *Transportation research part C: emerging technologies*, 40:143–159, 2014.

[56] Claudio Feliciani, Luca Crociani, Andrea Gorrini, Giuseppe Vizzari, Stefania Bandini, and Katsuhiro Nishinari. A simulation model for non-signalized pedestrian crosswalks based on evidence from on field observation. *Intelligenza Artificiale*, 11(2):117–138, 2017.

[57] Zijiang Zhu, Zhenlong Hu, Weihuang Dai, Hang Chen, and Zhihan Lv. Deep learning for autonomous vehicle and pedestrian interaction safety. *Safety science*, 145:105479, 2022.

[58] Raphael Korbmacher and Antoine Tordeux. Review of pedestrian trajectory prediction methods: Comparing deep learning and knowledge-based approaches. *IEEE Transactions on Intelligent Transportation Systems*, 23(12):24126–24144, 2022.

[59] Kaliprasana Muduli and Indrajit Ghosh. Predicting pedestrian-vehicle interaction severity at unsignalized intersections. *Traffic Injury Prevention*, 26(2):252–261, 2025.

[60] Muhammad Fahad, Zhuo Chen, and Yi Guo. Learning how pedestrians navigate: A deep inverse reinforcement learning approach. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 819–826. IEEE, 2018.

[61] Libo Sun, Jinfeng Zhai, and Wenhu Qin. Crowd navigation in an unknown and dynamic environment based on deep reinforcement learning. *IEEE Access*, 7:109544–109554, 2019.

[62] Zhanteng Xie and Philip Dames. Drl-vo: Learning to navigate through crowded dynamic scenes using velocity obstacles. *IEEE Transactions on Robotics*, 39(4):2700–2719, 2023.

[63] Cheng-En Cai, Sai-Keung Wong Wong, and Tzu-Yu Chen. Vehicle–pedestrian interaction for mixed traffic simulation. *Computer Animation and Virtual Worlds*, 36(3):e70031, 2025.

[64] Martin Treiber and Dirk Helbing. Microsimulations of freeway traffic including control measures. *at-Automatisierungstechnik*, 49(11):478, 2001.

[65] Chien-Yen Chang, T Hugh Woo, and Sen-Feng Wang. Analysis of pedestrian walking speeds at crosswalks in taiwan. *Journal of the Eastern Asia Society for Transportation Studies*, 9:1186–1200, 2011.

[66] Meng Zhang, Abdeljalil Abbas-Turki, Yazan Mualla, Abderrafiaa Koukam, and Xiaowei Tu. Coordination between connected automated vehicles and pedestrians to improve traffic safety and efficiency at industrial sites. *IEEE Access*, 10:68029–68041, 2022.

[67] Alexander Cui, Sergio Casas, Abbas Sadat, Renjie Liao, and Raquel Urtasun. Lookout: Diverse multi-future prediction and planning for self-driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16107–16116, 2021.

[68] Sander Schulhoff, Michael Ilie, Nishant Balepur, Konstantine Kahadze, Amanda Liu, Chenglei Si, Yinheng Li, Aayush Gupta, H Han, Sevien Schulhoff, et al. The prompt report: A systematic survey of prompting techniques. *arXiv preprint arXiv:2406.06608*, 5, 2024.

[69] Ioannis Karamouzas, Nick Sohre, Rahul Narain, and Stephen J Guy. Implicit crowds: Optimization integrator for robust crowd simulation. *ACM Transactions on Graphics (TOG)*, 36(4):1–13, 2017.

[70] ES Van Haeringen, Charlotte Gerritsen, and Koen V Hindriks. Emotion contagion in agent-based simulations of crowds: a systematic review. *Autonomous Agents and Multi-Agent Systems*, 37(1):6, 2023.

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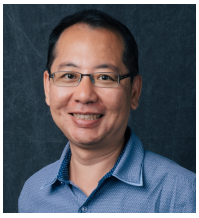
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